The Open Group Guide

The Agile Enterprise Architect Playbook
Version 1.0

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The Agile Enterprise Architect Playbook, Version 1.0

ISBN: 1-957866-05-5

Document Number: G226

Published by The Open Group, November 2022.

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Preface

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This Document

This document is The Open Group Guide: The Agile Enterprise Architect Playbook, Version 1.0. It has been developed and approved by The Open Group.

It is a living document. Additional Agile practices will be discussed in future releases.

The Agile practices discussed in this document were not chosen based on importance, but rather on interest by the contributing authors. Each Agile practice (or concept) will be discussed at a summary level with hyperlinks to sources with more detailed information. The content structure will be customized based on the topic. Where applicable, some topics will have real-life examples.

We hope you find this document helpful in your Agile journey.

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Acknowledgements

(Please note affiliations were current at the time of approval.)

The Open Group gratefully acknowledges the authors and the contribution of the following people in the development of this document:

- Alexsander Wyka, ATE Enterprise Ltd.
- Mark Dickson, The Open Group

The Open Group gratefully acknowledges the following reviewers who participated in the review of this document:

- Peter Ridgeway, Fujitsu
- Hari Krishnan Lakshminarayanan, Tata Consultancy Services Ltd.

Referenced Documents

The following documents are referenced in this guide.

(Please note that the links below are good at the time of writing but cannot be guaranteed for the future.)

Agile 2022	TOGAF® Series Guide: Enabling Enterprise Agility (G20F), published by The Open Group, April 2022; refer to: www.opengroup.org/library/g20f
ArchiMate 2022	How to Use the ArchiMate® Modeling Language to Support the TOGAF® Standard, The Open Group Guide (G21E), published by The Open Group, April 2022; refer to: www.opengroup.org/library/g21e
Arthur 1992	Rapid Evolutionary Development: Requirements, Prototyping, and Software Creation, Lowell Jay Arthur, published by John Wiley & Sons, 1992
Checkland 1990	Soft Systems Methodology in Action, Peter Checkland and Jim Scholes, published by John Wiley & Sons, 1990
Cockburn 2001	Agile Software Development, Alastair Cockburn, published by Addison Wesley, 2001
Cohn 2006	The Planning Onion: Making Sense of Planning in an Agile World, Mike Cohn, 2006
DeMarco 1982	Controlling Software Projects, Tom DeMarco, Yourdon Press, January 1982
DeMarco 2003	Waltzing With Bears: Managing Risk on Software Projects, Tom DeMarco, Dorset House, March 2003
DPBoK 2020	Digital Practitioner Body of Knowledge™ Standard (also known as the DPBoK™ Standard), a standard of The Open Group (C196), published by The Open Group, January 2020; refer to: www.opengroup.org/library/c196
Gilb 1988	Principles of Software Engineering Management, Tom Gilb, published by Addison Wesley, January 1988
Martin 1995	Great Transition: Using the Seven Disciplines of Enterprise Engineering, James Martin, published by Amazon, 1995
O-AA 2020	The Open Agile Architecture™ Standard (also known as the O-AA™ Standard), a standard of The Open Group (C208), published by The Open Group, September 2020; refer to: www.opengroup.org/library/c208
Sprints 2022	TOGAF® Series Guide: Applying the TOGAF ADM using Agile Sprints (G210), published by The Open Group, April 2022; refer to: www.opengroup.org/library/g210
TOGAF 2022	The TOGAF® Standard, 10th Edition, a standard of The Open Group (C220), published by The Open Group, April 2022; refer to: www.opengroup.org/library/c220
Weinberg 2011	Quality Software Management, Volume 1: Systems Thinking, Gerald M. Weinberg, published by Dorset House Publishing Co. Inc., US, February 2011

1 Introduction

1.1 Overview

This document is intended as a go-to-guide that shows how to use Agile practices with the TOGAF® Standard [TOGAF 2022]. It is intended to complement other TOGAF Guides by focusing on practical guidance from the authors' experiences.

This document is structured to discuss Agile practices applicable to each architecture level as identified in the TOGAF® Series Guide: Enabling Enterprise Agility [Agile 2022]:

- Strategic
- Segment
- Capability

Strategic will focus on Agile practices whose scope is enterprise-wide; for example:

- Agile planning
- What is the value of Agile practices?
- Agile Architecture Principles
- How is Agile enterprise-level governance implemented?

Segment will focus on Agile practices utilized at the program/portfolio level. These are:

- How do we handle innovation?
- Minimum Viable Architecture (MVA)
- Stakeholder collaboration
- Agile roles
- Agile budgeting
- Agile acquisition

Capability will focus on the Agile practices utilized to create a specific solution, such as;

- Agile documentation
- Pair architecting
- Agile team collaboration

1.2 Recommended Reading

This document is recommended to be read in conjunction with the following TOGAF® Series Guides:

Document	Summary
TOGAF® Series Guide: Enabling Enterprise Agility [Agile 2022]	This document describes in general terms how the TOGAF Standard can be adapted to support an "Agile enterprise". It is written to be applicable to any Agile delivery method that follows the commonly accepted Agile approach of iterative development through a series of sprints.
TOGAF® Series Guide: Applying the TOGAF ADM using Agile Sprints [Sprints 2022]	This document describes how sprints can be applied in a number of scenarios: • Collaboratively applying Enterprise Architecture and the TOGAF ADM following an Agile approach • Creating an Enterprise Architecture tuned for use together with Agile practices • Supporting and collaborating with Agile teams for a given business need • Enabling Enterprise Architects to leverage an Agile practice to assure that the Agile delivery of a solution is in alignment with strategic objectives

1.3 The Need for this Playbook

As an Enterprise Architect, you are required to evaluate the goals and objectives of your enterprise and identify a balanced and an integrated set of capabilities that then achieve those goals and objectives in an effective manner.

An enterprise has many different and competing goals and objectives and many different capabilities and solutions that it needs to create, integrate, and evolve. Enterprise Architecture provides a foundation for appropriate change within and between enterprises.

In some cases that will require ensuring that the solution meets many regulatory and legal aspects that must work on deployment. In other cases, the solution and the building blocks may have few constraints and can have variable but functional implementations that improve over time.

Enterprise Architecture evaluates the nature of the change before identifying the appropriate change for a specific change activity. Figure 1 shows characteristic change profiles against the amount of change likely to be encountered across a reasonably large enterprise. They reflect the different risk and integrity profiles to be considered before deciding on change lifecycles and practices.

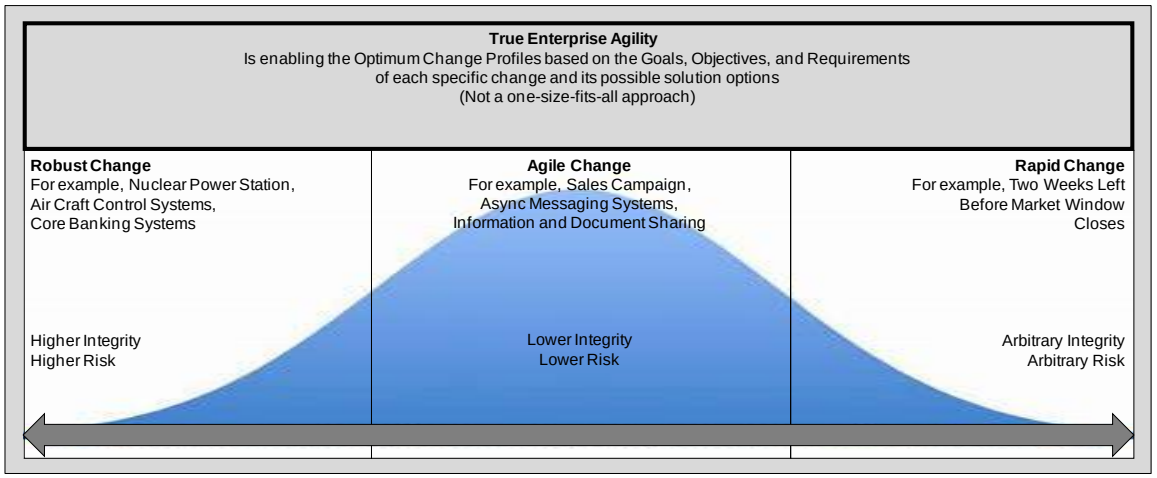

Figure 1: Enterprise Change Profiles

The most Agile enterprises are able to support all of the change profiles (based on specific change circumstances) and then use effective and relevant change practices. As can be seen, the general Agile practices, while applying generally in all three profiles, only apply specifically to the central type of change (which is the most frequently found). Understanding when and how to use these practices is the key to agility across the full spectrum of enterprise change. The following will be responsibilities of an Enterprise Architect:

- Identifying and facilitating the creation of a set of building blocks and patterns which can be implemented in operational business and information systems
- Providing guidance on how those building blocks, patterns, and systems can be created and evolved
- Assisting all other architects and designers in making decisions and resolving conflicts

All architecture work is a trade-off between:

- Requirements (including constraints)
- Possible solution components
- Solution integration
- Adoption of new technology/capabilities
- The approach to delivering change

Such trade-offs are at the heart of decisions made by architects.

This document provides a practical perspective (based on significant experience of using both Agile and architecture practices at scale) of how to use Agile practices effectively and appropriately and avoid the trap of continually failing fast and never really succeeding. It will also help in avoiding the analysis and design paralysis that delays otherwise valuable faster implementation and use.

2 Strategic Architecture Agile Practices

This chapter explores the Agile practices that are applied at the strategic level. We start at the high level, exploring what is Agile architecture, the value it provides to the enterprise, what Agile principles should be adopted, and how governance should be implemented.

2.1 Agile Planning

2.1.1 Ideas and Concepts

Agile practitioners describe architecture work as the important decisions you need to make. Agile approaches focus on planning and change activity, rather than the nature, structure, and composition of the components, sub-systems, and systems that make up the Architecture Landscape of the enterprise. A defining characteristic of Agile-style Architecture Landscape management is recognizing that all decisions do not need to be made upfront. Options should be preserved, and decisions delayed, until the last responsible moment.

The selection of approach should be one of an understanding of context and requirements. There will be some change that needs to be anticipation-led and some that needs to be adaptation-led. The challenge for both – to create an evolving enterprise's Architecture Landscape, and flexibility in the components, sub-systems, and systems within that landscape – is therefore that the process and the elements evolved by the process operate, anticipate, and adjust in line with the goals, objectives, and requirements of the enterprise.

When Enterprise, Segment, and Solution Architects work in heavily Agile-style organizations, this relationship between anticipation and adaptation is reflected in the following two classes of architecture:

- **Intentional Architecture**: expresses the desired effects of the architecture, consisting of a purposeful set of statements, models, and decisions representing some future architectural state

  International architecture also provides a set of planned architectural strategies and initiatives, which enhance product architecture quality attributes and promote inter-team design and implementation synchronization.

- **Emergent Architecture**: architecture elements that emerge from the solution or product delivery team during development, as higher-level patterns and principles are established from the development effort in progress

  Being derived from actual implementation, emergent architecture directly affects customer satisfaction and delight and provides practical technical evidence to reframe the intentional architecture guidelines in support of a fully evolutionary and iterative implementation approach.

In more Agile-focused organizations, it will often be assumed that emergent architect approaches are preferable to more heavily intentional ones. Consequently, more emphasis will be given to self-organizing teams and localized decision-making than organization shaping of teams and collaborative decision-making.

2.1.1.1 *Working With Agile Planning Ideas as TOGAF Enterprise Architects*

As mature Enterprise Architects, how should we respond to this approach and how can we work effectively with it? The simple answer is very easy because the TOGAF Standard and the associated framework already incorporate this approach.

As an open framework, the TOGAF Standard not only already includes the ideas behind Agile, but is also designed to be customizable and flexible, in response to any specific set of goals, objectives, requirements, and approaches.

2.1.2 Agile Planning with the TOGAF Standard

Agile planning began and is still significantly founded in the arena of software development. More recently, the long-running concept of business adaptability has been more closely linked to the concepts emerging from Agile planning. Both come together within the framework of Enterprise Architecture. A key element in this perspective is looking forwards based on indicators that sense future issues and opportunities, as well as understanding the current baseline and performance in the past. This drives the anticipation aspects embedded in Agile planning.

One of the key concepts behind Agile thinking is a problem identified formally by Barry Boehm back in 1981.[1] Boehm identified that when developing software, the discovery nature of the work meant that there was significant variation in the accuracy of estimates and solution sketches at the early stages of change. DeMarco in Controlling Software Projects [DeMarco 1982] and Waltzing For Bears [DeMarco 2003] confirmed these observations and promoted the concept of an estimating quality factor for change estimates. The variation associated with an estimating quality factor for any relatively complex project or larger size (bigger than a product developed by just a few people) is around a factor 2 – 4. You can be four times too large in your estimate just as you can be four times too small (although in practice the latter seems to be more common).

2.1.2.1 *The Cone of Uncertainty and the TOGAF Standard*

We can take this representation of the cone of uncertainty (that is familiar to Agile practitioners) and use it to describe how the TOGAF Standard provides the framework to integrate Agile approaches (it does of course also provide much more). In Figure 2, the Agile and TOGAF Standard terms for managing change cycles are shown.

Understanding the readiness of the organization to accept change, identifying the issues, and preparing to deal with them can reduce the uncertainty level facing the architecture development represented by the top side of the cone of uncertainty. Additional guidance on Business Transformation Readiness Assessments can be found in the TOGAF Standard. In the TOGAF Standard the change to be explored is called a *vision* and in Agile it is called a *product*. In both cases the success of the vision/product is in the implementation of a set of capabilities that delivers a set of outcomes satisfactory to the sponsors/users of the desired change. In both approaches the vision/product is always "aspirational" rather than actual. That is, it reflects the reality of the cone of uncertainty and is continually reviewed and, if needed; reframed and refactored in response to external environmental change, internal capability changes, and increasing levels of knowledge; as the solution comes closer to actual implementation.[2]

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[1] Refer to: https://en.wikipedia.org/wiki/Barry_Boehm.

[2] Of course, the degree of potential change depends on the nature of the change. Currently most information system solutions are combinations of existing components, re-used components, and new components. For change implementing well-known components, the aspiration will be close to the final implementation, and there may be little need for adaptation. When a totally novel solution is being created, such as new software to solve a new problem, then the aspiration may be very sketchy, and it may require the solution

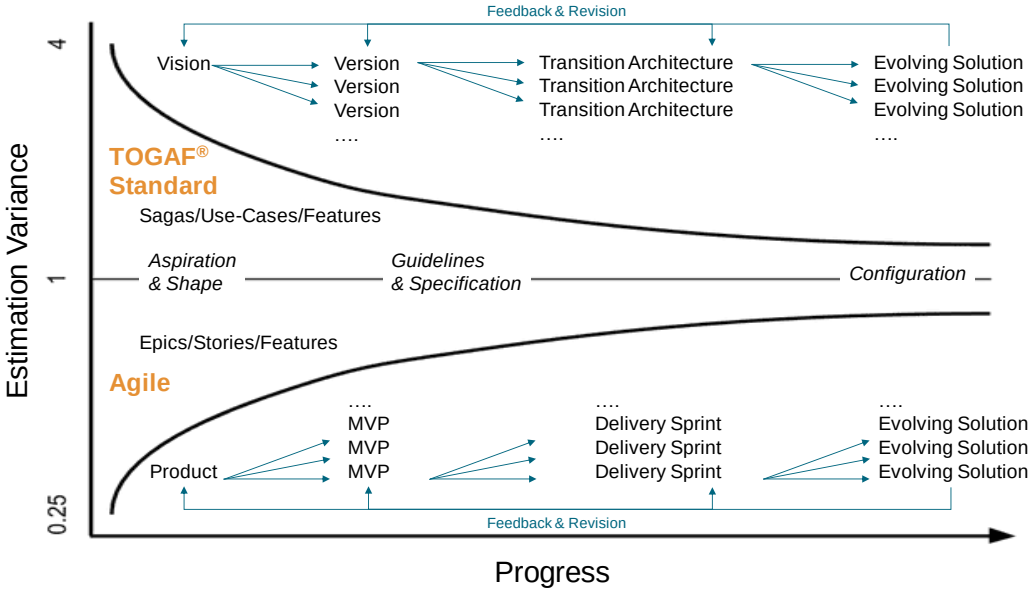

Figure 2: Aligning the TOGAF Standard and Agile Concepts

- Once the vision or product outline is approved, then it is broken down into a series of versions in the TOGAF Standard, and a series of Minimum Viable Product (MVP) units in Agile; these should consist of the smallest cohesive set of capabilities that would support effective business change for their users

- For each version or MVP (which can be developed in parallel), more detailed evaluation may be needed to be able to specify specific chunks of delivery that, while not providing the full cohesive set of capabilities in the version or MVP, do provide elements of value that can be tested, and may be able to be used by the business to provide more limited capability before the full set of capabilities is available

  In change that requires more control and direction, the specifications will be more detailed than change that only needs to know the outcomes that will be generated. In the TOGAF Standard approach this will determine the level of detail needed by the Architecture Building Blocks (ABBs).

- Each one of these sets of deliverable capabilities (called Transition Architectures in the TOGAF Standard and delivery sprints in Agile) is then prepared for re-use/acquisition/creation such that the implementation and operations teams can provide and deploy the solution

- The evolving solution (with the associated actual configuration) is then available to the business users to test, use, and assure that it meets their acceptance criteria and, in particular, delivers the agreed service levels

to be almost fully implemented through a series of increasingly capable prototypes before the final solution is fully known. Today most information systems change tends more towards the former re-use/existing component shape than the completely novel shape. However, there are many products and their deployments into customer environments that do have significant novelty (particularly new specialized software products).

This demonstrates how the Agile concepts, which are particularly strong as a change moves through the MVP and delivery sprint activities, align perfectly with the framework for change laid out in the TOGAF Standard.

The MVA becomes founded on the changes and their associated visions in the enterprise change management backlog, which contains all the suggested changes for that enterprise over the relevant period.

2.1.3 The Planning Onion And Agile Architecture

The Planning Onion

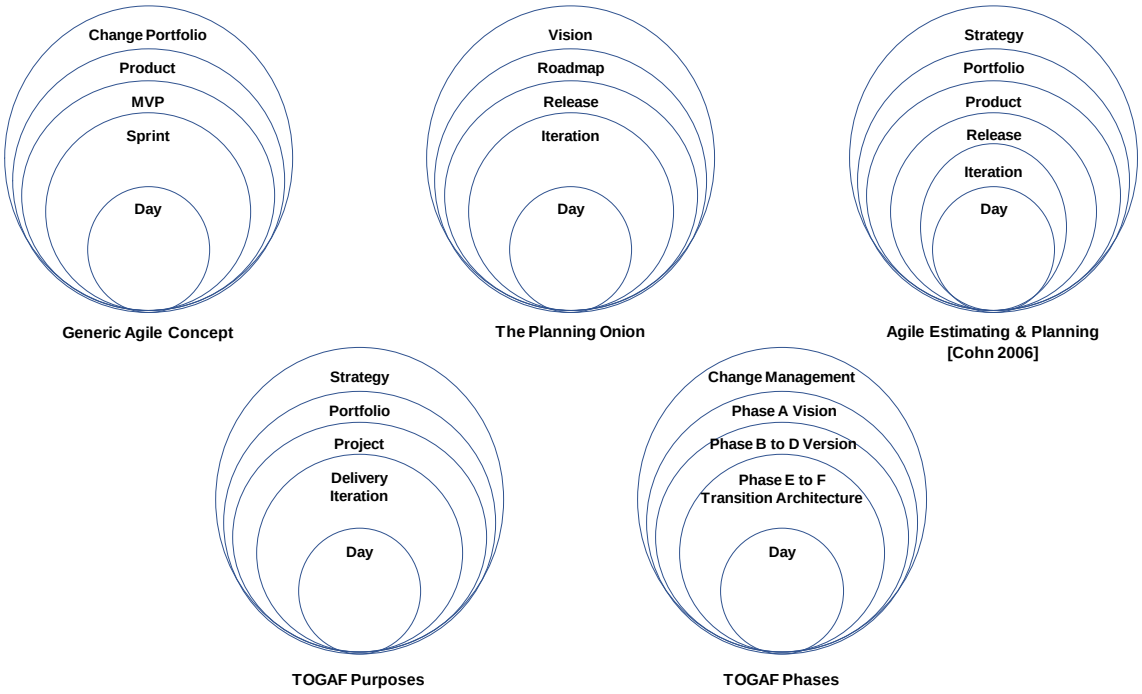

Figure 3: Mapping Multiple Viewpoints of the Planning Onion

The planning onion is a representation of the different levels of planning and execution needed when dealing with enterprise-level change. Figure 3 shows the representation of the planning onion from The Planning Onion: Making Sense of Planning in an Agile World [Cohn 2006] and its relationship to Agile approaches and the TOGAF Standard.

Any medium to large-scale enterprise needs to place Agile-style delivery techniques into a broader context of interoperability across many related contexts. This representation also demonstrates why Agile-at-scale approaches such as the Scaled Agile Framework® (SAFe®) have evolved, extending more common Agile techniques with those associated with the TOGAF Standard and more generalized enterprise-level scope.

2.1.3.1 Applying these Concepts when Using the TOGAF Standard

Changes affecting the two organizing and shaping purposes (Strategy and Portfolio) are reviewed by the Enterprise and Segment Architects to identify architectural issues and/or possibilities that can be progressed. It is the foundation for the main change roadmaps used for the evolution of the

broad Architecture Landscape. They are defined and managed by Enterprise and Segment Architects.

Changes associated with specific visions/products will have a lead Solution Architect developing those visions/products to the degree needed to gain stakeholder approval. The changes will then move through the common TOGAF Standard and Agile change elements of vision version/MVP and Transition Architecture/delivery sprint.

In this way, the TOGAF Standard provides the space for the broader evaluation of enterprise and segment-level goals and objectives and the development of the associated roadmaps and guidelines to govern broader interoperable change.

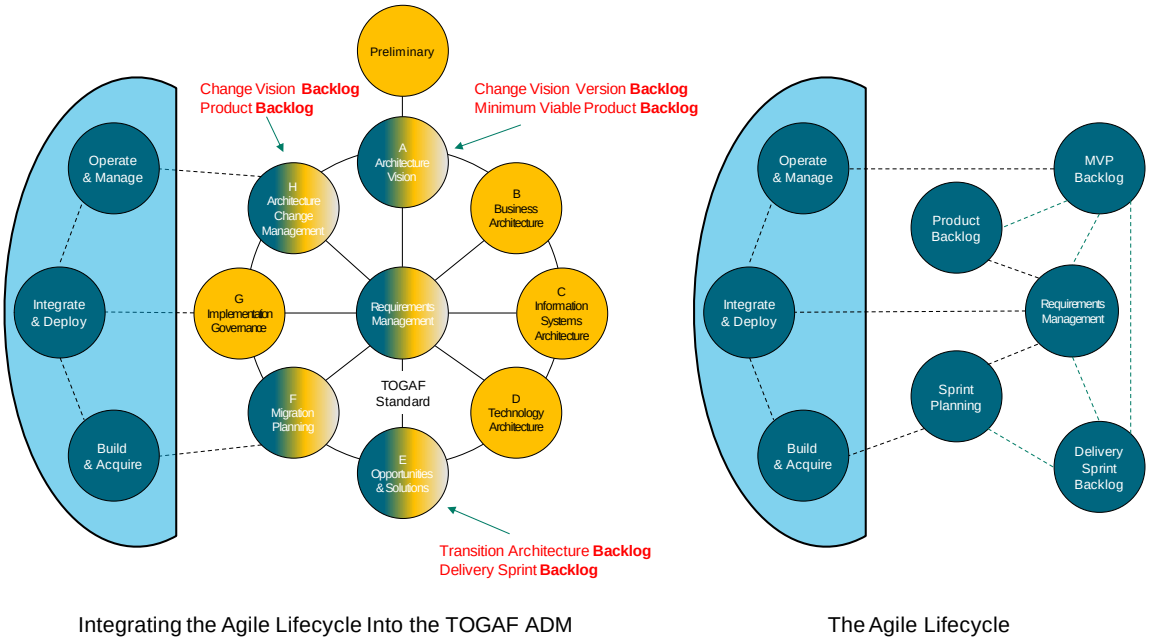

Figure 4: Integrating the TOGAF Standard and Agile – Agile Architecture Practice

Figure 4 shows how the major Agile elements align with the TOGAF Architecture Development Method (ADM); with both approaches working from the same set of backlogs for Architecture Landscape change. The TOGAF Standard generates specifications to be implemented by the build and acquire, integrate and deploy, and operate and manage aspects associated with the implementation of an Agile delivery sprint. Those specifications are defined by the ABBs and their organization into deliverable work packages to enable the creation of implementable Solution Building Blocks (SBBs) that will enable the execution of the changed business services.

This integration with Agile approaches emphasizes the potential for creating and evolving more adaptive and responsive components that will deliver specific capabilities and promote improved user and customer services; in addition to the specific implementation, feedback-driven elements of change that have been a continuous part of all professional information systems development approaches[3] and which have been variously incorporated into the TOGAF Standard approach.

[3] Such as:
Quality Function Deployment (QFD), developed in Japan beginning in 1966; refer to:
https://en.wikipedia.org/wiki/Quality_function_deployment
Principles of Software Engineering Management [Gilb 1988]

The higher-level elements of:

- Strategy – drivers, goals, and objectives
- Guidelines – principles, policies, and standards
- Domains – business, data, application, and technology
- Segments – portfolios, programs, and platforms
- Architecture patterns – modularity, coupling, and cohesion
- Service-orientation, etc.

are provided by the TOGAF Standard for consideration of how to manage large-scale complex Enterprise Architecture Landscapes, in addition to the fast delivery of new capabilities into live business usage. These are applied by Enterprise and Segment Architects when they develop reference models, blueprints, and guidelines and are used by Solution Architects when they are relevant to the specific solution change being addressed.

Phases B, C, and D of the TOGAF ADM provide the space where greater levels of intentional architecture can be incorporated into the solution delivery process when needed (reflected in the set of ABBs needed to support the change). In these phases, the first step is to choose the viewpoints, reference models, and tools that are of use in support of delivering a new and/or changed solution. There is no compulsion in choosing any of those; they are only needed when the solution requires more careful and/or extensive understanding of the nature and detail of a specification. Solutions being developed for changes associated with arenas such as nuclear power, aircraft control systems, complex health procedures, and auditable transaction control will often need agreed, shared, and approved specifications prior to both prototype and/or final implementation.

Blending these approaches together, combined with the skills and effective decision-making capabilities of architects, enables an enterprise to create and evolve an Architecture Landscape that has both the level of specificity it requires and the level of flexibility that it needs to exploit to remain competitive and relevant. Together they provide a broad foundation for a truly Agile architecture.

2.2 The Value of Agile Practices

Within the scope of this document, the value of Agile is defined as the results that will be realized when stakeholders are working together at the same pace and realizing the results that are satisfactory to their requirements.

Dynamic Systems Development Method (DSDM), first released in 1994; refer to:
https://en.wikipedia.org/wiki/Dynamic_systems_development_method
Objectory and object-oriented software engineering, created by Ivar Jacobsen, 1988-1993; refer to:
https://en.wikipedia.org/wiki/Objectory
Rapid Evolutionary Development: Requirements, Prototyping, and Software Creation [Arthur 1992]
Quality Software Management, Volume 1: Systems Thinking [Weinberg 2011]
Great Transition: Using the Seven Disciplines of Enterprise Engineering [Martin 1995]
Unified Modeling Language™ (UML®), originally designed by Grady Booch, James Rumbaugh, and Ivar Jacobson in 1995
Soft Systems Methodology in Action [Checkland 1990]
Agile Manifesto, 2002; refer to: https://agilemanifesto.org/display/000000002.html

These values:

1. Encourage putting people before processes
2. Enable frequent and regular delivery of results that will be released to the customer
3. Work in partnership with customers to realize results that are satisfactory
4. Enable adjustment of plans as needed

2.2.1 How Agile Value is Found within the TOGAF ADM

The TOGAF Standard is designed to complement any delivery style, including Agile.

Agile practices come with values that can be matched to the responsibilities of Enterprise Architects using the TOGAF Standard.

Below are the different TOGAF domains where Agile value can be measured:

- **Business Architecture**: the business strategy, governance, organization, and key business processes
- **Data Architecture**: the structure of an organization's logical and physical data assets and data management resources
- **Application Architecture**: a blueprint for the individual applications to be deployed, their interactions, and their relationships to the core business processes of the organization
- **Technology Architecture**: the logical software and hardware capabilities that are required to support the deployment of business, data, and application services; this includes IT infrastructure, middleware, networks, communications, processing, and standards

To test if value is realized, you can use the TOGAF ADM/TOGAF Standard/Agile Manifesto matrix in Table 1, which illustrates what Agile value is achieved in the different TOGAF domains at different phases of the ADM.

In the context of Enterprise Architecture "Working Software" is replaced by "2. Frequent and regular delivery of value/solution to the customer".

Table 1: Mapping of TOGAF ADM Phases to Domains

Agile Value / ADM Phase	1. Encourages putting people before processes	2. Frequent and regular delivery of value/solution to the customer	3. Collaborating with customers	4. Adjusting plans as needed
Phase A: Architecture Vision	√	√	√	√
Phase B: Business Architecture	√	√	√	√
Phase C: Information Systems Architecture	√	√	√	√

Agile Value / ADM Phase	1. Encourages putting people before processes	2. Frequent and regular delivery of value/solution to the customer	3. Collaborating with customers	4. Adjusting plans as needed
Phase D: Technology Architecture	√	√	√	√
Phase E: Opportunities & Solutions	√	√	√	√
Phase F: Migration Planning	√	√	√	√
Phase G: Implementation Governance	√	√	√	√
Phase H: Change Management	√	√	√	√

A common vision should always be in place before starting any request. Without a common vision, stakeholders will not achieve the value of Agile.

There is value in ensuring that all the different domains work cohesively as a whole and can be changed with agility and discipline without compromising the vision. This is an important responsibility of an Enterprise Architect.

The TOGAF ADM ensures that the "What" which is supposed to be delivered is not compromised and all the phases of the ADM will be addressed in Agile.

Stakeholders and their concerns are identified and addressed during Phase A: Architecture Vision.

2.3 Agile Architecture Principles

"Principles are general rules and guidelines, intended to be enduring and seldom amended, that inform and support the way in which an organization sets about fulfilling its mission.

Architecture Principles reflect a level of consensus across the enterprise, and embody the spirit and thinking of existing Enterprise Principles. Architecture Principles govern the architecture process, affecting the development, maintenance, and use of the Enterprise Architecture." [TOGAF 2022]

2.3.1 How the TOGAF Standard Supports Architecture Principles

The TOGAF Standard identifies a set of potential Architecture Principles for enterprises. The Agile Manifesto has 12 principles for general Agile-style change.

As an Enterprise Architect you are likely to be involved in setting the enterprise and domain/segment principles for your organization.

2.3.2 Developing and Using Architecture Principles

Architecture Principles create the highest level of direction for the shaping of change across your organization.

When developing Architecture Principles consider the balance between change and stability and novelty and risk needed by your organization.

Develop the set of Architecture Principles that will effectively shape your organization to achieve its goals and objectives. This set will usually be between 10 and 20 principles.

As part of that set consider some Agile Architecture Principles that direct the way that all solutions – at enterprise, segment, and/or capability level – are developed, implemented, and evolved.

Use the five additional Agile Architecture Principles suggested here as a starting point and, where needed, revise or customize them to the specific needs of your organization.

2.3.3 A Test of Good Principles

The foundation of success for good principles is that they are actually followed when change occurs and approximately 80 to 90% of that change can be implemented in a manner consistent with or close to them.

It is likely that a small but continually existing set of changes will challenge the principles and highlight issues or areas where significant decision-making is needed. The additional value of good principles is that they also provide a basis for good decision-making in these situations where their rationale and implications statements help to clarify how best to respond.

2.3.4 Additional Architecture Principles

In this document we have identified five additional Architecture Principles that you may consider when embedding Agile practices in your enterprise.

Name	Self-organizing teams
Statement	Support self-organizing teams, working with the right tools, within an appropriate context for collaboration and integration. Each team should have clear goals, a framework within which to create the relevant solution, and freedom to act as they deem appropriate; within the context of the degree of coupling and cohesion needed with other teams.
Rationale	Giving people freedom to act once they have the right information, tools, and expectations is the best approach for delivering meaningful and sustainable architectures and change.
Implications	Training and learning are crucial in preparing your people for effective action, decision-making, and accountability. The associated governance processes need to be proportionate, focused, and well understood at all levels.

Name	Communicate
Statement	Communicate and feed back at all levels of scale and composition. Ensure that you clearly identify all of your stakeholders. Communicate and collaborate effectively and tailor your deliverables so that they can be understood by the appropriate stakeholders.

Rationale	Despite having self-organizing teams with appropriate autonomy there are also many situations where multiple perspectives are needed to address difficult problems and interoperability.
Implications	Continually build relationships and shared understanding and utilize well-chosen architecture definition and management mechanisms for collaboration and discussion.

Name	Balance
Statement	Find the right blend of fail fast and right first time at all levels of architecture. Ensure that each change is evaluated against its specific circumstances and choose the right change lifecycle and approach to accuracy.
Rationale	Innovation, risk, and regulation need to be balanced in all change. You need to balance your organization's preference and the specific nature of each change at all levels of architecture.
Implications	In greater controlled and regulated change (greater focus on right first time) continually ensure that you are progressing with as frequent deliveries and as much learning as possible. In more innovative and faster change (greater focus on fail fast) ensure that you are not being slapdash, partial, and do not create difficult to escape *cul de sacs*.

Name	Openness
Statement	Promote open problem identification and open problem solving. Recognizing and discussing when directives (principles, policies, and standards) and/or requirements cannot easily be followed/addressed is at the heart of good Agile architecture practice.
Rationale	Architectural thinking is valuable when there are issues that cannot be easily solved. Prompting for and rewarding open discussion and addressing problems are the most important aspects of a successful architecture practice.
Implications	Train and reward architects to highlight and openly address issues, not to hide and minimize their impact. Real architecture agility is the ability to identify and address difficult issues/solutions more efficiently and effectively.

Name	Interoperability and Modularity
Statement	Create open, interoperable architectures based on appropriate modularity. Your architecture will need to evolve in most circumstances. The foundation for this is effective modularity that promotes localized use, integration, and re-use and repurposing of functionality over time.
Rationale	Architectures built for change (but with enough stability) enable organizations to respond faster to changing environments and requirements.

Implications	Build in enough capability to enable this change, based on the underlying concepts of coupling and cohesion between modules. Ensure that your teams are trained and that your building blocks are created to enable this Agile architecture in practice.

2.4 Enterprise-Level Architecture Governance

Enterprise-level Architecture Governance is the practice of ensuring that changes to the capabilities of an enterprise are developed to support the enterprise's strategy in terms of operational, financial, and reputational performance and are conducted in an effective, efficient, and proper manner.

It is a continuous process that provides an overview of the progress of change, the associated building blocks and deliverables, and their implemented impact on operations. Change projects should satisfy relevant requirements, rules, and regulations and operate within any specific guardrails or expectations placed upon that change.

Enterprise-level Architecture Governance identifies, reviews, and (as appropriate) enforces a series of directives that should be applied by change projects. These enable/constrain them to deliver solutions that address not only the specific requirements of that project but also the interoperability of the delivered solutions across the wider context of the overall enterprise, and within the desired timeframe.

2.4.1 How is Enterprise-Level Architecture Governance Applied in the TOGAF Standard?

Enterprises identify their strategic change perspective in terms of goals and objectives, and their specific change in terms of requirements.

- **Goals and Objectives** provide intent and/or direction of change
- **Requirements** provide statements of what is needed, together with associated criteria of success or acceptance

In achieving those goals and objectives and implementing the requirements, a set of directives identify how these should be approached. These cover:

- **Principles:** general expectations, rules, and guidelines
- **Policies:** how principles should be applied in different contexts
- **Standards:** specifics of things to which to conform
- **Best Practices:** specific ways of working

In the TOGAF Standard, Architecture Governance promotes, reviews, and records decisions made when new solutions are created and the degree to which they have followed the goals, objectives, requirements, and directives. As a result of such governance, solutions are identified as complying, requiring dispensations before continuing, or halted for further review. This ensures that change occurs in a relevant, effective, and traceable manner.

2.4.2 The Challenge of Enterprise-Level Governance of Agile Architectures

Agile practices emphasize the responsibility of each change project team to make relevant design and implementation decisions. This does not mean that there are no formal goals, objectives, requirements, or objectives; rather it means that as change progresses these elements should be considered in terms of not just what was originally identified, but also what has been discovered during development.

Most enterprises require change to occur in different ways for different types of solution, addressing the need for both more conservative risk-managed change as well as more speculative fast-to-market change. Good Enterprise Architecture Governance enables the optimum balance between these different approaches, clearly identifying when wider interoperability, rules, and regulations may constrain more speculative change and discovery, and *vice versa*.

Your governance should address both of these aspects and allow for reasoned decisions to be made as close as possible to the implementation context of each solution.

2.4.3 Working with Agile Teams

Your approach as an Enterprise Architect working with Agile teams is to lead, set up, and prepare the Segment and Solution Architecture, design and build teams to understand the overall context for their activities, and ensure that architecture and design decisions are shared where necessary. You should engage with the freedom and spirit of discovery emphasized in Agile practices, such that decisions about principles, policies, standards, and best practices are developed as collaboratively as possible. See also the TOGAF® Series Guide: Enabling Enterprise Agility §4.4.4 [Agile 2022].

The key to doing this well is to make sure that everyone has a clear understanding of the coupling and cohesion elements of all levels of change, and understands the general patterns of interoperability that have been identified across the enterprise for optimum capability use and re-use.

You should be open to review of decisions and directives based on feedback from individual projects, but any change that involves other than minimal coupling between solution components across multiple domains cannot be left purely to the individual change team.

2.4.4 Setting Expectations to Enable Success

To work with all teams effectively there is a need to ensure that:

- The architects, designers, and builders across the enterprise have been properly briefed and trained about the Enterprise Architecture, the nature of change, and the associated directives and guardrails
- Information about the Enterprise Architecture and the directives is easily available
- Design decisions are documented, with clear reasons and implications
- Everyone is clear about the governance process and the associated review points, and the basis on which changes are authorized to progress

2.4.5 Working with Agile Solution Architects, Designers, and Builders

Approach issues in an open manner, looking to generate agreed solutions from the many different stakeholders that may be involved. Do not start reviews with predefined ideas and solutions (unless there is a legal or regulatory reason why such ideas and solutions have to be applied); rather, clearly frame the issues and invite proposals from all. If possible, help to shape the review towards agreed solutions that respect many different viewpoints.

You still have to lead, and you may have to make a final decision, but utilize the skills and capabilities of all in making your decisions.

2.4.6 Responsibility for Overall Architecture Effectiveness

As an Enterprise Architect you are responsible to senior management and potentially the Governing Board for the effective development and implementation of capabilities that will deliver their strategy and provide effective day-to-day operations.

Agile practices emphasize that there are many contributors to doing this effectively and that knowledge and experience at the “sharp end” are crucial to creating the right blend of planned and emergent structure and solutions.

Your main responsibility is to understand the interaction between these ways of delivering change and operations, and to lead across the full scope of the enterprise, addressing issues and conflicts as they emerge.

In any enterprise of scale and complexity you will not have the view of the individual Agile change team that can focus on the smaller and more local issues; you have to address the broader and integrated view. Enterprise Architecture Governance is the ultimate oversight of the implementation of this enterprise viewpoint.

3 Segment Architecture Agile Practices

In this chapter, we will explore Agile practices at the segment level. Some of these practices can be considered strategic. It is very subjective.

3.1 Innovation Dialog

There are many techniques to help innovation, such as Design Thinking, Fish Bowl, 6-3-5 Brainwriting, Open Space Technology (OST), and World Café.[4] Design thinking is used within many Agile frameworks. It follows a three-step process of inspiration, ideation, and implementation. The inspiration step focuses on gaining a deep understanding of the problem from the customer's point of view, to produce a clear definition of the problem to be solved. During ideation, different potential solutions are identified, which leads to the final step of implementation in which prototypes are developed and tested. See also the TOGAF® Series Guide: Enabling Enterprise Agility §5.2.1 [Agile 2022].

The first step, inspiration, is the most important. A variety of techniques are commonly used to explore the problem and understand it deeply from the customer's point-of-view. "Storming" events such as event storming and body storming are intended to provide as close to a personal experience as possible, and direct observation or document research can also provide insight. Market research and analyst reports can indicate market directions, competitor activities, and emerging technologies. These can provoke ideas for novel solutions.

3.1.1 Innovation Dialog and Enterprise Architecture

Innovation dialog is needed throughout the architecture lifecycle to help ensure that the best solutions are delivered to the business. Innovation can occur at any stage of the lifecycle, for example:

- **Vision**: innovative commercial approaches, such as contract structures that release more value
- **Business Architecture**: innovative business processes, such as short-term rentals of spare capacity in some asset to boost utilization
- **Technology Architecture**: innovative technology solutions, such as augmented reality views of machinery to assist with maintenance

3.1.2 Support for Innovation in the TOGAF ADM

The TOGAF ADM is intended to be customized to fit the local environment, and innovation can be applied in any phase. In particular, the standard specification covers innovation in Phase D (Technology Architecture) and Phase H (Architecture Change Management). Within Phase D, the potential to incorporate emerging technologies should be examined as a way to produce novel

[4] Design Thinking: https://en.wikipedia.org/wiki/Design_thinking;
Fish Bowl: https://en.wikipedia.org/wiki/Fishbowl_(conversation);
6-3-5 Brainwriting: https://en.wikipedia.org/wiki/6-3-5_Brainwriting;
Open Space Technology (OST): https://en.wikipedia.org/wiki/Open_Space_Technology;
World Café: https://en.wikipedia.org/wiki/World_caf%C3%A9_(conversation).

solutions. Within Phase H, the potential for innovation in business process or technology is recognized as a driver for change.

Adopting new and innovative business practices or technology will often require considerable change, so an assessment of the readiness of an organization to transform can be useful. The TOGAF Standard includes a Business Transformation Readiness Assessment process, which enables a detailed assessment of an organization's readiness to enact a business transformation. It examines factors such as the desire and need for change, funding availability, sponsorship, and ability to execute. The assessment could be a precursor to a particularly disruptive business transformation, where the change necessary is considerable.

3.2 Minimum Viable Architecture

The Minimum Viable Architecture (MVA) is just enough architecture to fulfill the purpose for which architecture is created. The MVA is built on the intentional architecture. The intentional architecture, as defined by the O-AA™ Standard [O-AA 2020], is a purposeful set of statements, models, and decisions that represent some future architectural state. Basically, you can think of the intentional architecture as a foundation.

The need for an MVA is acknowledgement that you do not know what you do not know. The architecture is kept modular and flexible. As you acquire firmer knowledge of the functional and non-functional requirements through the sprints, you enhance and optimize your architecture.

3.2.1 How is the MVA Applied in the TOGAF Standard?

The TOGAF Standard supports the MVA pattern. Since the order, iteration, and components of the TOGAF Standard are defined by the enterprise, the scope can be aligned to a Transition Architecture. It is a continuous architecture process that is a way to define, build, and refactor the architecture of a system of systems to reflect the current understanding of the business need and whether the current solution is fit-for-purpose.

Rather than having the Architecture Definition Document defined before the starting development of the first major release, break it up into many short internal releases framed by less frequent external releases. Each minor release contains additional or modified ABBs and SBBs. Some building blocks will be deployed but not released since they are required for future function. As requirements are better understood, you may actually need to deactivate a building block.

3.2.2 Using an MVA in the Real World

We developed an eCommerce Software as a Service (SaaS) product which enabled customers to develop and host eCommerce sites with their own brands. The technology was brand new and needed to be integrated with legacy products and refactor code from a previous failed project.

Because the user experience was not defined and needed to be validated through focus groups, we created many small releases using an MVA approach. With each small release we were able to get user feedback that helped us to define a marketable product. Given that the product was based on modular services, we were able to add or modify the architecture at each release. We did first create an intentional architecture for the foundation.

A challenge with this approach was that user interface standards frequently changed, resulting in high recoding.

3.3 Stakeholder Collaboration

When collaborating with stakeholders, you focus on two types of communication: work information and architecture.

Information radiators are used to provide transparency regarding the work being performed. They provide a view of the team's daily progress, work quality, impediments, and risks. Effective information radiators should be simple, unambiguous, current, transient, influential, highly visible, and minimal in number.

Architecture views should be adapted to the stakeholders with the help of the stakeholders. As Figure 5 shows, face-to-face communication will give the best results. Always understand that your stakeholders have their own stakeholders to whom they will report. If your stakeholders understand your architecture, they will explain and sell your architecture to their stakeholders.

The DPBoK™ Standard, §6.4.3.2.3 (Modeling and Visualization) [DPBoK 2020] goes into detail about visualization.

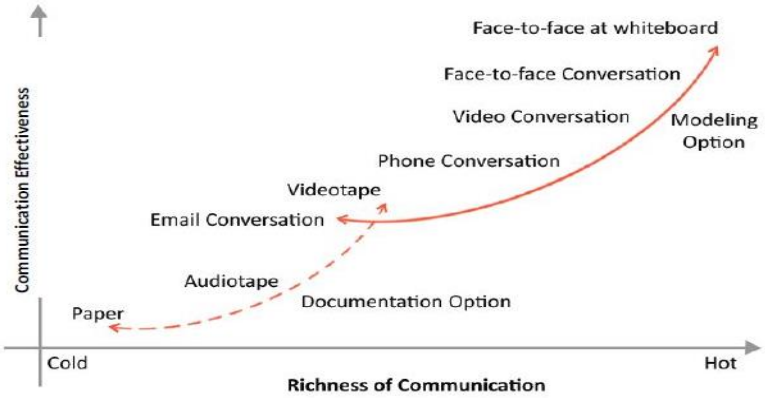

Figure 5: Stakeholder Communication Model[5]

3.3.1 Communicating with Stakeholders about Architecture

When you read about stakeholders in most literature and training, it is about how to manage the stakeholders. There are a few recurring patterns on how to identify and classify your stakeholders. You can find detailed examples of this in the TOGAF Standard – ADM Techniques, §3 (Stakeholder Management) [TOGAF 2022].

But in an Agile context you don't manage stakeholders, you collaborate with them. It is the outside-in focus instead of the inside-out, where you try to control the stakeholders. Enterprise Architecture stakeholders including external customers need to become involved across the entire architecture lifecycle.

Think about how you can involve the stakeholders in your architecture work. You have to ensure that stakeholders are involved in not just the presentations and reviews of the results of each iteration, but also in decision-making across the entire product lifecycle.

[5] Copyright © 2001 Alistair Cockburn [Cockburn 2001].

They are the customers of your architecture and the implementation of it. Think and talk about the business value; not just financial, but other types of value such as ease of change, effective and happy staff, delighted customers. Different values will attract different types of stakeholders getting them involved instead of managed. You can get so much valuable important information from them if you manage to involve and understand them. Let the stakeholders collaborate with each other and you and together you will find the correct path forward.

3.3.2 Agile Stakeholder Collaboration in the TOGAF ADM

Preliminary Phase

When building an Enterprise Architecture Capability, try to focus on external drivers. This applies to (among others):

- Principles – try to formulate Architecture Principles which reflect customer value; do not focus on internal effectiveness before identification of business priorities, otherwise you will stay driven by technology
- Metamodel – while choosing what to model, try to choose those entities that give you the best possibility to model customer value, customer interaction, and market
- Scope of the architecture – do not try to model everything
- Guides – create guides explaining effective ways of communicating architecture

Architecture Vision

While formulating the vision, concentrate on customer value. At the vision level, do not try to deal with details. Allow it to happen later – the vision should be a starting point for further action and the means of communicating the intent. Try to make your vision simple and straightforward.

Use the TOGAF Stakeholder Management method to identify stakeholders. Ensure that the outside-in view is considered and demand that the stakeholders are involved.

Remaining Phases

Ensure that the architecture is communicated in all possible ways to ensure all stakeholders are involved and understand. Define good artifacts ensuring that they communicate the architecture to the stakeholders instead of creating perfectly correct models.

3.4 Agile Roles

This section focuses on how roles in an Agile environment are supported for Enterprise Architecture and management.

Enterprise Architecture is a discipline for proactively defining a conceptual blueprint to enable businesses to respond to disruptive forces. Enterprise Architecture roles extend to deliver the Segment Architect role and Solution Architect roles in cases of skill shortage or when the Enterprise Architecture portfolio is still under development.

The following are Enterprise Architecture roles:

- To understand the Baseline and Target Architecture
- To understand the business environment
- To deliver architectural designs
- To document the Architecture Landscape and Technical Reference Model (TRM), specific business problems, system or sub-system level such as management and security (more IT-centered)
- To increase re-usability, whereby data and technology are shared across the enterprise
- Ownership of records of decisions in the Enterprise Architecture repository
- To set and govern the implementation of architectural standards to govern and maintain the IT architecture

The roles above still apply and are supported by the TOGAF Standard; however, they become Agile Enterprise Architecture roles when done in sprints:

- Focus less on big models, supporting all domains
- Focus on up-front planning
- Promote just-in-time rapid collaboration on the deliverables
- Ensure Enterprise Architecture Governance
- Identify relevant teams and stakeholders to deliver the solution(s)
- Drive rapid decision-making and ownership of decisions
- Drive requirements communication amongst stakeholders

Note: The above are not Enterprise Architecture jobs, but Agile teams thrive on the idea of being multi-disciplinarians.

3.4.1 Enterprise Architecture Agile Roles

3.4.1.1 How the TOGAF ADM Supports the Enterprise Architect Role

The TOGAF Standard is designed to complement any deliverable by all roles: Enterprise Architects, Segment Architects, and Solution Architects.

Agile Enterprise Architecture is supported by the TOGAF ADM by allowing any phase to be executed concurrently by the Agile development team.

The TOGAF Standard and the Enterprise Architect make sure that the long-term strategies and goals of the enterprise are understood by the Agile development team.

The architect has a responsibility for ensuring the completeness (fitness-for-purpose) of the architecture, in terms of adequately addressing all the pertinent concerns of its stakeholders; and the integrity of the architecture, in terms of connecting all the various views to each other, satisfactorily reconciling the conflicting concerns of different stakeholders, and showing the trade-offs made in so doing (as between security and performance, for example).

The choice of which particular architecture views to develop is one of the key decisions that the Enterprise Architect has to make. The choice has to be constrained by considerations of practicality, and by the principle of fitness-for-purpose (i.e., the architecture should be developed only to the point at which it is fit-for-purpose, and not reiterated *ad infinitum* as an academic exercise).

3.4.1.2 Test of Success

To test if value is realized you can use the TOGAF ADM/TOGAF Standard/Agile Manifesto matrix in Table 1:

- Stakeholders and their concerns are identified and addressed during Phase A: Architecture Vision
- Business requirements are ensured to be aligned to the strategy
- Needed decisions are identified and resolved at the right time; costly mistakes can be avoided in the early stages
- Facilitated communication of gaps and risks
- Insight is gained from implementation of the decisions made, promoting collective learning
- Cross-skilling of policies, process, and standards between teams
- Commitment to quality and speed is driven by all stakeholders in the team
- Change management is addressed during the delivery of the solution
- Strong work relationships and trust are built amongst the team
- Repository of information, surveys, adjusted models, and use of Agile methods in all steps of the framework
- Solutions are delivered quickly and are fully tested and accepted by all stakeholders
- Continuous improvement of solutions during reviews and design lifetime resulting from repetitive architectural reviews

3.4.2 Agile Management Roles

There are a number of management roles to be aware of in an Agile organization, particularly in a hybrid Agile environment. Below are some examples, together with a proposed simple definition of each.

- The **Portfolio Manager** works with groups of programs, separate projects, and other activities aimed to achieve one strategic goal
- The **Program Manager** works with several related projects that pursue a bigger common goal

  They focuses on higher-level management of these projects. This includes interdependencies, risks, the economics of the whole program, as well as the work required to coordinate the work of several Project Managers.

- The **Product Manager** identifies the customer need and the larger business objectives that will articulate what success looks like for a product
- The **Project Manager** leads one or several separate projects responsible for cost
- The **Product Owner** is a member of an Agile product team responsible for defining user stories and prioritizing the backlog, ensuring these are understood by other team members while maintaining the conceptual integrity of the features or components for the delivery team
- The **Scrum Master** is involved in the day-to-day processes of the product development process; they oversee the implementation of the Scrum process

3.4.2.1 Agile Management Roles In Enterprise Architecture

So, how are these roles used in an Enterprise Architecture setting?

The Portfolio Manager is supported by the Enterprise Architecture team, which creates the Strategic and Segment Architecture defining potential future roadmaps.

The Program Manager is supported by the Enterprise Architecture team, which creates the just-in-time Segment and Capability Architecture for scaled Agile development.

The Product Manager is supported by the Enterprise Architecture team creating product Strategic Architectures.

The Project Manager is supported by the Enterprise Architecture team creating just-in-time Capability and Transition Architectures including Implementation and Migration Plans.

The Product Owner is supported by the Enterprise Architecture team inserting work packages and user stories (business/application/technology services) into the product and sprint backlogs.

The Scrum Master interacts with the chief Enterprise Architect combining the Enterprise Architecture framework with the Agile development framework.

3.4.3 Focus on Collaboration in the TOGAF ADM

Preliminary Phase

The (Scrum of) Scrum Master and chief Enterprise Architect collaborate to define the interaction between the Enterprise Architect and Agile development teams.

Phase A (Architecture Vision)

The Product Manager and Enterprise Architecture team interact to create the vision of the product.

Phases B to E (Architecting Phases)

The Product Manager, Product Owner, and Enterprise Architecture team interact to define the architecture of the product.

Phase E (Opportunities & Solutions)

The Project Manager, Product Owner, and the Enterprise Architecture team interact to define the work products and the Architecture Roadmaps.

Phase F (Migration Planning)

The Scrum Manager, Project Manager, and the Enterprise Architecture team interact to confirm management framework interactions for the Implementation and Migration Plan.

The Project Manager and the Enterprise Architecture team interact to define the Implementation and Migration Plan, risks, and business values of work packages.

3.5 Agile Acquisition

Agile acquisition is the strategy, structure, and process to effectively adopt and apply Agile software development methodologies. Agile acquisition integrates planning, design, development, and testing into an iterative lifecycle to deliver small, frequent, incremental capabilities to an end user.

Agile acquisition requires integrated customer and contractor processes and partnerships. It embraces the Agile development principles of self-organizing cross-functional teams, frequent capability deliveries, continuous stakeholder participation and feedback, and evolving requirements.

3.5.1 Agile Acquisition Best Practices

Best practices for Agile acquisition include:

- Understand why and how you want to use Agile acquisition
- Understand that there will be a major need for constant communication and real-time collaboration
- Focus on outcome-based solutions instead of specification-based
- Develop best practices using external and internal collaboration
- Partner with the right supplier

3.5.2 Enterprise Architecture Role in Agile Acquisition

What is the role of Enterprise Architecture and architects in an Agile acquisition? Just as in any other acquisition, Enterprise Architecture is important. The Enterprise Architect does not only look at the acquisition, but also how that acquisition will fit into the larger picture of the entire business and IT landscape (the context) of the purchaser and the long-term relations with the provider.

Trust

The most important aspect of Agile acquisition is the trust between the two partners: the purchaser and the provider. Enterprise Architecture can be part of creating that trust from different perspectives:

- Ensure that the provider understands the environment within the purchaser; the business, IT, and finance environment, including their culture and Agile experiences
- Ensure that the purchaser understands the environment within the provider; the business, IT, and finance environment, including their culture and Agile experiences
- Develop an evolving outcome-based incremental Enterprise Architecture in collaboration with the supplier architects based on strategic objectives and goals and qualities
- Make sure the purchaser and the supplier can deliver the needed hard decisions just in time

Governance/Support

Govern/support the implementation together with the supplier architects:

- Ensure stakeholder participation (purchaser and supplier) in demos, retrospectives, and sprint planning and even daily scrum meetings
- Ensure that the purchaser has the final say in decisions and ensure they are well analyzed and agreed

3.5.3 Experience from Fixed Price

In a large, fixed-price telco product the supplier allowed the purchaser to have the final say. The purchaser wanted to change the architecture and after long discussion won the argument. A short time later, the purchaser discovered that the architecture he pushed would cause more problems than it solved so it was changed back, but at a large cost.

From a trust perspective, the purchaser did not trust the supplier architecture, and we probably should have provided better rationale and explanation.

3.6 Rolling Forecast Budgeting

More and more organizations are choosing to use rolling forecasts for their financial control. A rolling forecast provides better opportunities to make the right financial decisions at the right time than with traditional planning. A rolling forecast means that the planning horizon is successively moved forward, while at the same time it is kept within limits.

3.6.1 Why Rolling Forecasts?

Accuracy

A common complaint about traditional annual budgeting is that by the time it is completed, it is already irrelevant. Rolling forecasts allow you to make quick tweaks along the way rather than

letting mistakes mount up and only giving yourself one opportunity to make those changes annually.

Agility

Rolling forecasts allow you to adjust the forecast to accommodate recent changes or trends, meaning you are able to respond better to time-sensitive decisions. Because your outlook is updated continuously, you will always have long-term data available when your organization needs to make an important business decision.

Driver-Based

Historical data is used to predict future numbers. And metrics like category growth, market share, human capital, and customer satisfaction are fed into your system. Any fluctuations to operational activities can be accounted for throughout the year, instead of just once.

3.6.2 Rolling Forecasts and Enterprise Architecture

Enterprise Architecture is an important stakeholder in the budgeting of an organization. The Transition Architectures and MVAs created are constantly input to the rolling budget.

In an Agile environment, the architectures and solutions are created incrementally with a constantly improved understanding of their costs and business values. This enables the organization to continuously maintain their rolling budget based on the best information available.

Each Transition Architecture consists of roadmaps, work packages, migration plans, and business cases for that transition which fits perfectly into the rolling forecast and also the portfolio management of the new and existing initiatives and their budgets, giving a more granular budget and planning understanding due to the incremental approach.

A rolling forecast can be equated to the refactoring concept often used in an Agile development.

3.6.3 Focus on Budgeting in the TOGAF ADM

Phase A (Architecture Vision)

- Develop the business case for the architectures and changes required
- Produce the value proposition

Phase E (Opportunities & Solutions)

- Formulate the Implementation and Migration Strategy
- Identify and group major work packages
- Identify Transition Architectures
- Create the Architecture Roadmap and Implementation and Migration Plan

Migration Planning

- Prioritize the migration projects through the conduct of a cost/benefit assessment

4 Capability Architecture Agile Practices

Most people when using Agile practices focus at the solution, product, or project level. This was the original scope for which Agile was designed. This chapter explores some of these practices. We reiterate that this categorization of Agile practices is subjective.

4.1 Collaboration Between Agile Teams

Many Agile practices focus on optimizing collaboration between Agile teams. In this section we explore these practices. For more detail information, refer to the TOGAF® Series Guide: Applying the TOGAF ADM using Agile Sprints [Sprints 2022].

4.1.1 Agile Practices to Support Scrum

Scrum is a very popular Agile framework for developing, delivering, and sustaining products in a complex or simple environment.

4.1.1.1 Sprint

A sprint (also known as an *iteration* or *timebox*) is the basic unit of development in Scrum. A sprint is a timeboxed effort; that is, the length is agreed and fixed in advance for each sprint. Each sprint aims to deliver valuable insights addressing stakeholders' concerns.

4.1.1.2 Sprint Zero

The goal of sprint zero is to come together to develop a minimal number of user stories, a project skeleton, and story mapping to develop a workable product. Sprint zero usually takes place just before or at the start of the product development effort.

4.1.1.3 Backlogs

A backlog is an ever-evolving list of requirements, prioritized by the stakeholders, that conveys to an Agile team which requirement to handle first. There are two types of backlog in any effort: the backlog for the entire effort and the backlog for each sprint.

4.1.1.4 Demonstration, Retrospective, and Planning

At the end of each sprint the sprint team will conduct three activities: a demonstration (demo) of what they have done, a retrospective analysis of sprint performance, and planning for the next sprint. The demos show the outcomes of the finished sprint to the stakeholders and the other sprint teams.

4.1.2 Scrum in the TOGAF Standard

The Scrum practices fit well with the TOGAF Standard. The concept of iterations has been around in the TOGAF Standard for many years. Together with the concept of partitions, they mimic the concepts of sprints and MVAs.

The architecture backlog for the entire effort is defined during Phase A (Architecture Vision) while creating the Statement of Architecture Work. The sprint backlogs are a statement of work for each partition (MVA). Architecture reviews (demos) are performed across the entire effort. At the end

of each partition (sprint), the TOGAF Standard states that there should be a lesson learned (retrospective) session.

4.1.3 Enterprise Architecture Sprints

Most Enterprise Architecture efforts can be broken down into a set of sprints (iterations).

4.1.4 Sprint Zero (Strategic Sprint)

In the context of Enterprise Architecture, it is recommended to start with a preparation sprint. Depending on the context it can be, for example, initial business needs, major capabilities, value streams, or high-level roadmaps for a business transformation. Inputs to this sprint are the high-level requirements for business and IT improvement, together with existing higher-level architectures.

Sprint zero is very similar to Phase A (Architecture Vision) of the TOGAF ADM, where the Enterprise Architecture effort backlog (i.e., the rationale, scope, and vision for the entire architecture) is defined in collaboration with all stakeholders. The stakeholders and their concerns are identified and collaboration plans with stakeholders are defined. The organization's readiness for change is investigated.

The input to sprint zero can be an architecture at a higher level; e.g., a strategic-level architecture is input to the segment-level sprint zero, and a segment-level architecture is input to a capability-level architecture.

4.1.5 Architecture Backlogs

Like Scrum, there are two types of backlog in architecture work: backlogs for the entire architecture and the backlog for each ADM sprint. The full architecture backlog is defined during sprint zero, but is constantly revised across the entire architecture effort. The architecture sprint backlog defines the TOGAF partition that is created during a sprint or set of sprints.

4.1.6 Review, Retrospective, and Planning

At the end of each sprint, the architecture is explained to and reviewed by the stakeholders as suggested in the TOGAF Standard. A retrospective as in Phase F (Migration Planning) of the TOGAF ADM is performed on the sprint work to ensure effectiveness and quality in the architecture work.

At last, the sprint backlog for the next iteration is defined from the full architecture backlog.

4.1.7 Sprinting the ADM

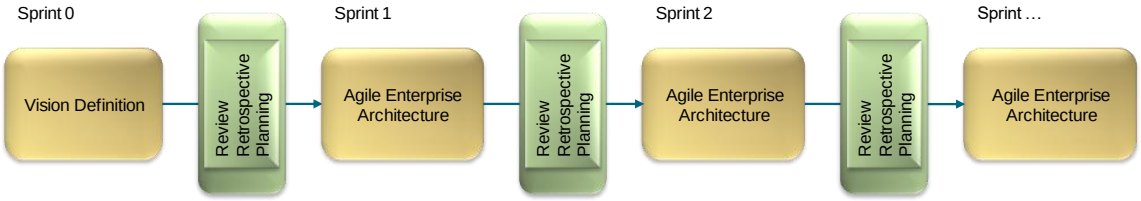

Figure 6: Enterprise Architecture Sprints

4.1.8 Collaboration Between Agile Teams

In an Agile enterprise, the Architecture team does not work in isolation. It collaborates with other Agile teams within the enterprise in different ways. Sometimes, the output from a sprint is an input to the sprint backlog of another team; sometimes people with different competencies collaborate in the same sprint team. It depends on the culture and Agile maturity of the enterprise.

4.1.9 Collaboration Between TOGAF and Agile Solution Development

The TOGAF architecture can be developed on its own, using Agile practices like Scrum, but often the Enterprise Architecture team needs to collaborate with other Agile teams. The most common collaborating partners are the Agile solution development teams.

The Agile Enterprise Architecture team creates MVAs using one or more sprints. These MVAs are the input to the Agile solution development backlogs.

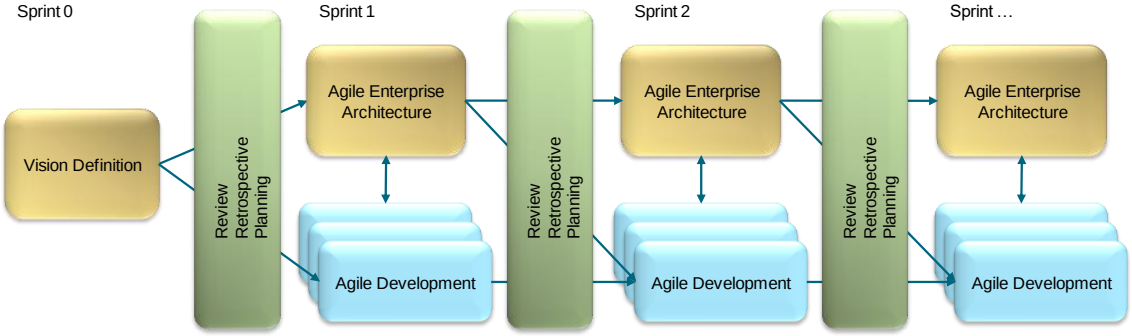

Figure 7: Enterprise Architecture and Agile Development Sprints

This will create positive effects:

- The business change stakeholders will have an earlier understanding of what they will receive from the demos
- Agile solution teams involved in creating and implementing the Enterprise Architecture ensures better architecture
- Agile solution teams' decisions and concerns are fed back to architects quickly
- Agile solution teams involved in creating and implementing the Enterprise Architecture ensures enhanced business value
- Agile solution teams understand the effects of Enterprise Architecture decisions quickly and can give feedback

4.1.10 Collaboration Between Enterprise Architecture and Agile Business Development

Collaboration with the Agile business development teams is just as important as we are moving to Agile enterprises. The Agile business development team is creating Minimum Viable Business Development Descriptions (MVBD). The MVBDs are input to the backlog of the Agile Enterprise Architecture.

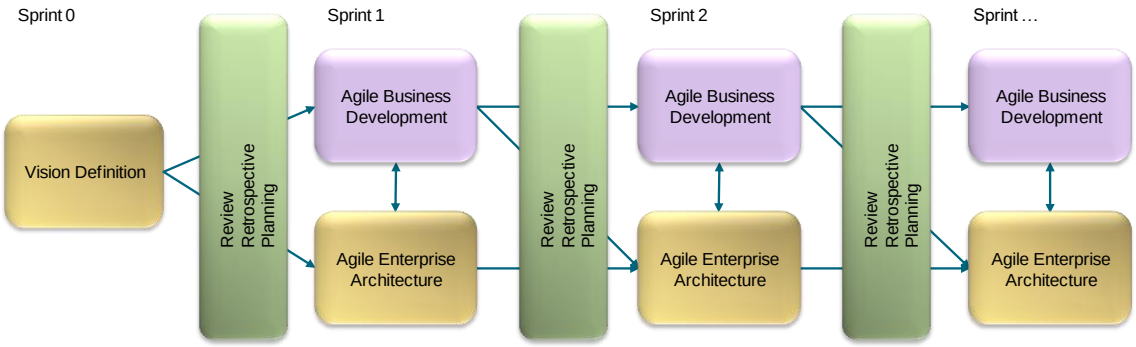

Figure 8: Enterprise Architecture and Business Development Sprints

This will create positive effects:

- Business development is involved in creating the business change
- Business development understands the effects of business decisions quickly
- Business development decisions are evaluated from a holistic perspective
- Business development proposals can be reworked with better understanding of the effects
- Architects can influence the business decisions, requirements, and development

4.1.11 Cross-Functional Teams

Cross-functional agility means that each sprint team consists of business development, architecture, and Agile solution development competencies. These competencies are collaborating in the same team. There are no specific teams, just mixed cross-functional self-organizing Agile teams. All improvement work is constantly managed through the product and sprint backlogs.

This will create positive effects:

- Closes access to all the skills necessary to effectively deliver value to customers
- Brings new insights to the team to define creative solutions and enhance development
- Breaks stereotypes and spurs innovative ideas though diversity
- Enables a common corporate language
- Resolves issues of conflicting priorities and improves conflict resolution
- Cross-organization alignment

4.2 Agile Documentation for Architecture

Architecture documentation is primarily a means to record and communicate architecture definitions. It can include words, pictures, diagrams, models, and other representations of the architecture. However, documents can also be other things; for example, a medium for reviews, deliverables against which payments are made, and sometimes a pile of paper ignored soon after it is produced.

One of the four fundamental values in the Agile Manifesto addresses documentation: "*[We value…] Working software over comprehensive documentation*", but also acknowledges that "*there is value in the items on the right*"; i.e., documentation. The history page of the Agile Manifesto goes on to say: "*We embrace documentation, but not hundreds of pages of never-maintained and rarely-used tomes*".

Your Agile documentation must balance the need to record and communicate architecture definitions with the need to enable rapid and incremental change. The effort invested in creating and maintaining documentation must be outweighed by the benefit of reading and using it. Guidance within the TOGAF Standard recognizes the need to select the appropriate viewpoints and tools, etc. that are appropriate to the needs of the endeavor, and this is consistent with the Agile principle of always doing "just enough, but no more". See also the TOGAF® Series Guide: Enabling Enterprise Agility §5.4.2 [Agile 2022].

4.2.1 Support of the Agile Practice in the TOGAF ADM

The TOGAF ADM describes many artifacts (catalogs, matrices, and diagrams) that define various aspects of architecture. These may be created by a variety of word processor or presentation tools, diagramming tools, or modeling tools. The TOGAF Standard – Architecture Content, §4 (Architecture Deliverables) [TOGAF 2022] describes the deliverables containing these artifacts, most of which would manifest themselves as documents that are generated during the architecture project.

The ADM phase descriptions specify which artifacts and deliverables should be input and output. However, the Standard does not specify how the artifacts and deliverables should materialize in your documents; for example, there is no specification of any particular standard document format or standard template.[6][7] Therefore, the format can be chosen to suit the local environment; for example, an architecture toolset or a DevOps toolchain.

4.2.2 Agile Documentation in Practice

You should document only the minimum content necessary to enable the next step. This means:

- Documentation is produced incrementally, and with an expectation of change

  Early increments should focus on short statements of vision and principles, and diagrams showing high-level structures and relationships.

- For some steps, the minimum necessary may be zero

- Towards the end of an architecture project, the "minimum content necessary" will be whatever is necessary in the future development and operation of the solution; this will not be zero

- Documentation tools and formats:

  — Give preference to tools and formats which enable fine-grain collaboration; for example, documents in markup formats held within a code management system, and tools supporting multiple simultaneous editors

[6] There are example templates within the TOGAF Library, but these are not part of the TOGAF Standard.
[7] Guidance on modeling TOGAF artifacts in the ArchiMate modeling language can be found in The Open Group Guide: How to Use the ArchiMate® Modeling Language to Support the TOGAF® Standard [ArchiMate 2022].

— Automation features such as approval workflows can increase efficiency and reduce cycle times

— The scale and complexity of an endeavor can influence the choice of tools; for example, for small-scale work, commodity office tools and shared folders may be sufficient; however, for large projects, purpose-made modeling tools may be beneficial

Accessibility over long periods of time should also be considered, which can become an issue for artifacts stored within a toolset repository. Toolsets naturally evolve and change over time, and there is a risk that old content can become inaccessible. One solution has been to export documentation to long-established file formats such as PDF.

4.3 Pair Architecting

Pair architecting is an Agile technique where the driver develops the code and the observer reviews the code. The two programmers switch roles frequently.

Architecting in pairs, following the pair program technique, will improve consistency, accelerate delivery, and improve the quality of the architecture model since mistakes are discovered earlier and there is an early consensus on the architecture.

While reviewing the architecture models created, the observer also considers the strategic direction of the architecture, coming up with ideas for improvement and likely future problems to address. This frees the driver to focus all of their attention on the modeling aspects of the current task, using the observer as a safety net and guide.

This allows the modeler to be an expert modeler and the observer to be a fellow modeler, topic expert, or a decision-maker.

Pair architecting also allows the participants to be at separate locations which is very beneficial when the team is geographically spread or topic experts/decision-makers are scarce.

Dual modeling moves editing responsibility between the participants.

Experience shows that discovering errors in the work early mitigates the cost of pair architecting. Other benefits are knowledge transfer and learning from each other.

4.3.1 Pair Architecting in the TOGAF ADM

Preliminary Phase

Pair architecting ways of working are defined. Define pairing roles and responsibilities.

Phases A to F and Architecture Requirements Management

Define the TOGAF artifacts in pairs where the benefits are high; e.g., create common understanding, get expert help, make decisions.

Phase G (Implementation Governance)

Work in pairs to handle support and compliance reviews to ensure a consistent message.

4.3.2 Pair Architecting Example

When creating the Security Architecture for a product, we collaborated in work sessions on a model connecting the ISO/IEC 27001/2[8] way of working to the necessary processes, IT components, and services. The work sessions were done both face-to-face using a large screen for collaboration and remotely using conference tools from home. One of us would manage the model and we would switch roles depending on the expertise of the participants.

[8] ISO/IEC 27001: Information Security Management; ISO/IEC 27002: Information Technology – Security Techniques – Code of Practice for Information Security Controls.

5 Final Thoughts

This document provides some Agile practices used by the authors while implementing projects with the TOGAF Standard. Since this is a living document, additional practices will be added in future releases based on new Agile delivery experiences.

If you have experience implementing Agile practices with the TOGAF Standard, we encourage you to join The Open Group and contribute your knowledge to the community.

Acronyms & Abbreviations

ABB	Architecture Building Block
ADM	Architecture Development Method
DSDM	Dynamic Systems Development Method
DXP	Digital Experience
MVA	Minimum Viable Architecture
MVBD	Minimum Viable Business Development Descriptions
MVP	Minimum Viable Product
O-AA	Open Agile Architecture
OST	Open Space Technology
QFD	Quality Function Deployment
SaaS	Software as a Service
SAFe	Scaled Agile Framework
SBB	Solution Building Block
TRM	Technical Reference Model
UML	Unified Modeling Language

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